

Load the dataset into a DataFrame df_ipl and display first 5 rows.

```
In [28]: import pandas as pd
df= pd.read_csv(r"C:\Users\VANSH\Downloads\IPL2025Batters.csv")
df.head()
```

Out[28]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s	6s
0	Sai Sudharsan	GT	759	15	15	1	108*	54.21	486	156.17	1	6	88	21
1	Surya Kumar Yadav	MI	717	16	16	5	73*	65.18	427	167.91	0	5	69	38
2	Virat Kohli	RCB	657	15	15	3	73*	54.75	454	144.71	0	8	66	19
3	Shubman Gill	GT	650	15	15	2	93*	50.00	417	155.87	0	6	62	24
4	Mitchell Marsh	LSG	627	13	13	0	117	48.23	383	163.70	1	6	56	37



Display shape and column names of the dataset.

```
In [11]: print(df.shape)
print(df.columns)
```

(156, 14)
 Index(['Player Name', 'Team', 'Runs', 'Matches', 'Inn', 'No', 'HS', 'AVG',
 'BF', 'SR', '100s', '50s', '4s', '6s'],
 dtype='object')

Display only "Player Name" and "Runs" columns.

```
In [14]: df[['Player Name', 'Runs']]
```

Out[14]:

	Player Name	Runs
0	Sai Sudharsan	759
1	Surya Kumar Yadav	717
2	Virat Kohli	657
3	Shubman Gill	650
4	Mitchell Marsh	627
...	...	...
151	Khaleel Ahmed	1
152	Mohit Sharma	1
153	Donovan Ferreira	1
154	Digvesh Singh	1
155	Manoj Bhandage	1

156 rows × 2 columns

Display data types using info().

In [15]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 156 entries, 0 to 155
Data columns (total 14 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Player Name     156 non-null   object
1   Team            156 non-null   object
2   Runs            156 non-null   int64
3   Matches         156 non-null   int64
4   Inn             156 non-null   int64
5   No              156 non-null   int64
6   HS              156 non-null   object
7   AVG             156 non-null   object
8   BF              156 non-null   int64
9   SR              156 non-null   float64
10  100s            156 non-null   int64
11  50s             156 non-null   int64
12  4s              156 non-null   int64
13  6s              156 non-null   int64
dtypes: float64(1), int64(9), object(4)
memory usage: 17.2+ KB

```

Select players who scored more than 500 runs.

In [23]: df[df['Runs']>500]

Out[23]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s	
0	Sai Sudharsan	GT	759		15	15	1	108*	54.21	486	156.17	1	6	88
1	Surya Kumar Yadav	MI	717		16	16	5	73*	65.18	427	167.91	0	5	69
2	Virat Kohli	RCB	657		15	15	3	73*	54.75	454	144.71	0	8	66
3	Shubman Gill	GT	650		15	15	2	93*	50.00	417	155.87	0	6	62
4	Mitchell Marsh	LSG	627		13	13	0	117	48.23	383	163.70	1	6	56
5	Shreyas Iyer	PBKS	604		17	17	5	97*	50.33	345	175.07	0	6	43
6	Yashasvi Jaiswal	RR	559		14	14	1	75	43.00	350	159.71	0	6	60
7	Prabhsimran Singh	PBKS	549		17	17	0	91	32.29	342	160.52	0	4	56
8	K L Rahul	DC	539		13	13	3	112*	53.90	360	149.72	1	3	52
9	Jos Buttler	GT	538		14	13	4	97*	59.78	330	163.03	0	5	52
10	Nicholas Pooran	LSG	524		14	14	2	87*	43.67	267	196.25	0	5	45

Select "Player Name" and "SR" where SR > 150 using loc.

```
In [30]: df.loc[df['SR']>150,['Player Name','SR']]
```

Out[30]:

	Player Name	SR
0	Sai Sudharsan	156.17
1	Surya Kumar Yadav	167.91
3	Shubman Gill	155.87
4	Mitchell Marsh	163.70
5	Shreyas Iyer	175.07
6	Yashasvi Jaiswal	159.71
7	Prabhsimran Singh	160.52
9	Jos Buttler	163.03
10	Nicholas Pooran	196.25
11	Heinrich Klaasen	172.69
12	Priyansh Arya	179.24
14	Abhishek Sharma	193.39
16	Phil Salt	175.98
17	Riyan Parag	166.52
19	Ryan Rickelton	150.97
20	Travis Head	162.60
23	Ishan Kishan	152.58
24	Shashank Singh	153.50
26	Dhruv Jurel	156.33
31	Tristan Stubbs	150.75
33	Sherfane Rutherford	157.29
35	Josh Inglis	162.57
37	Axar Patel	157.48
38	Jitesh Sharma	176.35
39	Vaibhav Suryavanshi	206.55
40	Naman Dhir	182.60
41	Devdutt Padikkal	150.60
42	Sunil Narine	170.83
43	Ayush Mhatre	188.97
45	Aniket Verma	166.19

	Player Name	SR
47	Dewald Brevis	180.00
48	Hardik Pandya	163.50
49	Nitish Rana	161.94
50	Rinku Singh	153.73
51	Ashutosh Sharma	160.62
53	Karun Nair	172.17
56	Tim David	185.14
58	Shahrukh Khan	179.00
59	Andre Russell	163.72
60	Abdul Samad	176.34
61	Marcus Stoinis	186.04
65	Vipraj Nigam	179.74
67	Washington Sundar	166.25
68	Ruturaj Gaikwad	150.61
69	Sameer Rizvi	153.16
74	Shubham Dubey	160.60
75	Rahul Tewatia	167.79
76	Pat Cummins	167.24
80	Jonny Bairstow	184.78
84	Romario Shepherd	291.66
85	Urvil Patel	212.50
86	Jacob Bethell	171.79
103	Avesh Khan	175.00
105	Kusal Mendis	200.00
108	Gerald Coetzee	154.54
109	Jamie Overton	214.28
124	Kwena Maphaka	400.00
125	Raj Angad Bawa	200.00
129	Akash Deep	300.00
137	Lockie Ferguson	400.00

Select players from team "RCB".

In [31]: `df[df['Team']=='RCB']`

Out[31]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s
2	Virat Kohli	RCB	657	15	15	3	73*	54.75	454	144.71	0	8	66
16	Phil Salt	RCB	403	13	13	1	65	33.58	229	175.98	0	4	48
28	Rajat Patidar	RCB	312	15	14	1	64	24.00	217	143.77	0	2	25
38	Jitesh Sharma	RCB	261	15	11	4	85*	37.29	148	176.35	0	1	24
41	Devdutt Padikkal	RCB	247	10	10	1	61	27.44	164	150.60	0	2	21
56	Tim David	RCB	187	12	9	6	50*	62.33	101	185.14	0	1	16
72	Liam Livingstone	RCB	112	10	8	1	54	16.00	84	133.33	0	1	4
73	Krunal Pandya	RCB	109	15	7	1	73*	18.17	86	126.74	0	1	9
77	Mayank Agarwal	RCB	95	4	4	1	41*	31.67	64	148.43	0	0	10
84	Romario Shepherd	RCB	70	8	3	1	53*	35.00	24	291.66	0	1	5
86	Jacob Bethell	RCB	67	2	2	0	55	33.50	39	171.79	0	1	9
112	Bhuvneshwar Kumar	RCB	14	14	6	3	8	4.67	25	56.00	0	0	1
138	Yash Dayal	RCB	4	15	3	1	3	2.00	8	50.00	0	0	0
155	Manoj Bhandage	RCB	1	1	1	0	1	1.00	4	25.00	0	0	0



Select players where Runs > 600 and display only "Player Name" and "Team".

In [32]: `df.loc[df['Runs']>600,['Player Name','Team']]`

Out[32]:

	Player Name	Team
0	Sai Sudharsan	GT
1	Surya Kumar Yadav	MI
2	Virat Kohli	RCB
3	Shubman Gill	GT
4	Mitchell Marsh	LSG
5	Shreyas Iyer	PBKS

Select players where 50s >= 5.

```
In [34]: df[df['50s']>=5]
```

Out[34]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s	6s
0	Sai Sudharsan	GT	759	15	15	1	108*	54.21	486	156.17	1	6	88	2
1	Surya Kumar Yadav	MI	717	16	16	5	73*	65.18	427	167.91	0	5	69	3
2	Virat Kohli	RCB	657	15	15	3	73*	54.75	454	144.71	0	8	66	1
3	Shubman Gill	GT	650	15	15	2	93*	50.00	417	155.87	0	6	62	2
4	Mitchell Marsh	LSG	627	13	13	0	117	48.23	383	163.70	1	6	56	3
5	Shreyas Iyer	PBKS	604	17	17	5	97*	50.33	345	175.07	0	6	43	3
6	Yashasvi Jaiswal	RR	559	14	14	1	75	43.00	350	159.71	0	6	60	2
9	Jos Buttler	GT	538	14	13	4	97*	59.78	330	163.03	0	5	52	2
10	Nicholas Pooran	LSG	524	14	14	2	87*	43.67	267	196.25	0	5	45	4
13	Aiden Markram	LSG	445	13	13	0	66	34.23	299	148.82	0	5	38	2

Select players with at least one century (100s > 0).

```
In [37]: df[df['100s']>0]
```

Out[37]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s	
0	Sai Sudharsan	GT	759		15	15	1	108*	54.21	486	156.17	1	6	88
4	Mitchell Marsh	LSG	627		13	13	0	117	48.23	383	163.70	1	6	56
8	K L Rahul	DC	539		13	13	3	112*	53.90	360	149.72	1	3	52
11	Heinrich Klaasen	SRH	487		13	13	2	105*	44.27	282	172.69	1	1	42
12	Priyansh Arya	PBKS	475		17	17	0	103	27.94	265	179.24	1	2	55
14	Abhishek Sharma	SRH	439		14	13	0	141	33.77	227	193.39	1	2	46
23	Ishan Kishan	SRH	354		14	13	3	106*	35.40	232	152.58	1	1	33
36	Rishabh Pant	LSG	269		14	13	2	118*	24.45	202	133.16	1	1	23
39	Vaibhav Suryavanshi	RR	252		7	7	0	101	36.00	122	206.55	1	1	18



Select players where Runs > 500 AND SR > 140.

```
In [40]: df[(df['Runs']>500)&(df['SR']>140)]
```


Out[40]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s
0	Sai Sudharsan	GT	759	15	15	1	108*	54.21	486	156.17	1	6	88
1	Surya Kumar Yadav	MI	717	16	16	5	73*	65.18	427	167.91	0	5	69
2	Virat Kohli	RCB	657	15	15	3	73*	54.75	454	144.71	0	8	66
3	Shubman Gill	GT	650	15	15	2	93*	50.00	417	155.87	0	6	62
4	Mitchell Marsh	LSG	627	13	13	0	117	48.23	383	163.70	1	6	56
5	Shreyas Iyer	PBKS	604	17	17	5	97*	50.33	345	175.07	0	6	43
6	Yashasvi Jaiswal	RR	559	14	14	1	75	43.00	350	159.71	0	6	60
7	Prabhsimran Singh	PBKS	549	17	17	0	91	32.29	342	160.52	0	4	56
8	K L Rahul	DC	539	13	13	3	112*	53.90	360	149.72	1	3	52
9	Jos Buttler	GT	538	14	13	4	97*	59.78	330	163.03	0	5	52
10	Nicholas Pooran	LSG	524	14	14	2	87*	43.67	267	196.25	0	5	45

Select players where Team = "MI" AND Runs > 400.

```
In [41]: df[(df['Team']=='MI')&(df['Runs']>400)]
```

Out[41]:

	Player Name	Team	Runs	Matches	Inn	No	HS	AVG	BF	SR	100s	50s	4s	6s
1	Surya Kumar Yadav	MI	717	16	16	5	73*	65.18	427	167.91	0	5	69	38
15	Rohit Sharma	MI	418	15	15	1	81	29.86	280	149.28	0	4	41	22

Sort players by Runs in descending order.

```
In [ ]:
```

Difference between df["Runs"] and df[["Runs"]]

```
In [42]: df["Runs"]    # It is a Series. (one square bracket)
```

```
Out[42]: 0      759
         1      717
         2      657
         3      650
         4      627
         ...
        151      1
        152      1
        153      1
        154      1
        155      1
        Name: Runs, Length: 156, dtype: int64
```

```
In [46]: df[["Runs"]]  # It is a DataFrame. (double square bracket)
```

```
Out[46]:
```

	Runs
0	759
1	717
2	657
3	650
4	627
...	...
151	1
152	1
153	1
154	1
155	1

156 rows × 1 columns

Why use & instead of 'and'?

```
In [47]: # It is because Pandas/Numpy built on C++ programming language, & helps in applying
```

```
In [ ]:
```